

TABLE IX
Logic Functions

S.No.	Decimal equivalent	Binary code	Logic function
0	1	0001	NOR
1	3	0011	NOT A
2	5	0101	NOT B
3	6	0110	EX-OR
4	7	0111	NAND
5	8	1000	AND
6	9	1001	EX-NOR
7	10	1010	BUFFER B
8	12	1100	BUFFER A
9	14	1110	OR

Other conversions are not used

Program design

Before starting the coding, one should know two components—dectobin and mux—for decimal to binary and multiplexer, respectively. These components can be used along with the 'port map' clause available in VHDL.

```
--decimal to binary encoder begins
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity dectobin is
port (decs:in integer;hexs :out
std_logic_vector(3 downto 0));
end entity dectobin;
architecture arcdtobin of dectobin is
begin
process (decs)
begin
case decs is
when 1 => hexs<="0001";--NOR
when 3 => hexs<="0011";-- NOT A
when 5 => hexs<="0101";-- NOT B
when 6 => hexs<="0110";-- EX-OR
when 7 => hexs<="0111";-- NAND
when 8 => hexs<="1000";-- AND
when 9 => hexs<="1001";-- EX-NOR
when 10 => hexs<="1010";-- BUFFER B
when 12 => hexs<="1100";-- BUFFER A
when 14 => hexs<="1110";--OR
when others => hexs <= "0000";
end case;
end process;
end architecture arcdtobin;
--decimal to binary encoder ends
```

The entity and architecture codes using mux component are as under:

```
--component multiplexer code begins
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity mux is
port(s1:in std_logic; s0:in std_logic;
i3_0 :in std_logic_vector(3 downto 0);muxy : out std_logic:= '0');
end mux;
```

```
architecture archmux of mux is
begin
process (s1,s0,i3_0)
begin
muxy<= (i3_0(0) and (not s1) and
(not s0)) or (i3_0(1) and (not s1)
and s0) or (i3_0(2) and s1 and (not
s0)) or (i3_0(3) and s1 and s0) ;
end process;
end archmux;
--multiplexer code ends
```

The entity and architecture codes for the main module of universal logic board using the above components are:

```
--universal board code begins
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity unilogic is
port (A:in std_logic; B:in std_logic;
decs:in integer:=0;y : out std_logic:= '0');
end unilogic;

architecture Behavioral of unilogic is
signal decouts:std_logic_vector
{3 downto 0};

component dectobin is
port (decs:in integer;hexs :out std_logic_vector(3 downto 0));
end component dectobin;

component mux is
port (s1:in std_logic; s0:in std_logic;
i3_0 :in std_logic_vector(3 downto 0);
```

```
muxy : out std_logic:= '0');
end component mux;

begin
DECH: dectobin port
map (decs,decouts);
MUX : mux port map (a,b,decouts,y);
end Behavioral;

--universal board code ends
```

Xilinx VHDL 9.2i has a separate test bench for designing and testing the code. From the default test bench, one has to make a few changes in the test bench code for various functions shown in bold below:

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.std_logic_unsigned.all;
USE ieee.numeric_std.ALL;

ENTITY unitb_vhd IS
END unitb_vhd;

ARCHITECTURE behavior OF unitb_vhd IS

--Component Declaration for the Unit
Under Test (UUT)
COMPONENT unilogic
PORT (
A : IN std_logic;
B : IN std_logic;
decs : IN integer;
y : OUT std_logic
);
END COMPONENT;

--Inputs
SIGNAL A: std_logic := '0';
SIGNAL B: std_logic := '0';
SIGNAL decins: integer:= 0;

--Outputs
SIGNAL y : std_logic;

BEGIN
-- Instantiate the Unit
Under Test (UUT)
uut: unilogic PORT MAP(
A => A,
B => B,
decins => decins,
y => y
);
```